

Generation Interconnection System Impact Study Report

For Queue Project Q19-1236

Sandhills 100 kV

June 02, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1236. The Interconnection Customer proposes a 124.5 MW solar photovoltaic facility interconnecting at the Sandhills 100 kV substation in the Ohio area. The request was received on July 02, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 493 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,694,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HATCH / OVERLOOK 100kV	BARAGA VLG / QUEEN CRK SUBSTATION 500kV	122.5	126	9.1
OVERLOOK / CEDAR BAYOU 100kV Line	CARO / MIAMI 100kV	103.0	107	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HATCH / OVERLOOK 100kV	Rebuild line with higher-capacity conductor	\$1,855,000	37% (\$689,000)
CEDAR BAYOU / OVERLOOK 100kV	Reconductor line	\$2,597,000	39% (\$1,005,000)
	Total allocated to project		\$1,694,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.